

Chapter -1

Computer Network and Communication

BROADBAND

Broadband is a high-speed, always-on internet connection that can transmit large amounts of data quickly.

1. WHAT IS BROADBAND?

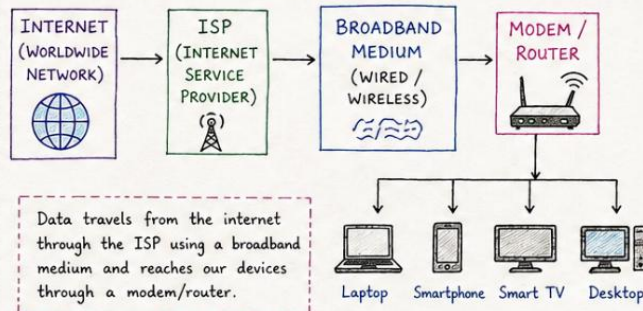
It is a wideband transmission technology that uses a wide range of frequencies to carry data, voice, video and other multimedia services simultaneously.

KEY FEATURES

- High speed
- Always-on connection
- Carries multiple signals at once
- Supports internet, TV and phone services together



2. HOW BROADBAND WORKS



4. BROADBAND vs DIAL-UP

BROADBAND	DIAL-UP
High speed	Low speed
Always-on (no dialing)	Connects only when you dial
Can transmit data, voice, video together	Transmits data only
Uses wide range of frequencies	Uses narrow range of frequencies
Suitable for streaming, gaming, video calls, etc.	Not suitable for high bandwidth applications

5. BROADBAND BANDWIDTH (ANALOGY)

Bandwidth is like the width of a road.

Narrow road (Narrow bandwidth)



→ Less data at a time → Slow

Wide road (Broadband)



→ More data at a time
→ Fast

More bandwidth = More data = Higher speed

3. TYPES OF BROADBAND

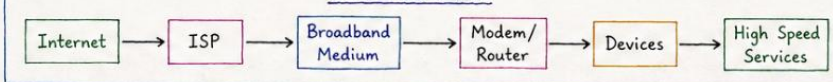
- ① DSL (Digital Subscriber Line)
 - Uses existing telephone lines.
 - Speed: moderate.
- ② CABLE BROADBAND
 - Uses TV cable lines.
 - Provides high speed.
- ③ FIBER OPTIC BROADBAND
 - Uses optical fiber (light signals).
 - Fastest, highly reliable.
- ④ WIRELESS BROADBAND
 - Uses radio waves (Wi-Fi, 4G/5G).
 - No cables required.
- ⑤ SATELLITE BROADBAND
 - Uses satellite signals.
 - Useful in remote areas.

6. APPLICATIONS OF BROADBAND

- High speed internet browsing
- Online streaming (YouTube, Netflix)
- Video conferencing
- Online gaming
- Cloud computing
- Smart home / IoT devices.
- E-learning, Telemedicine, etc.



QUICK SUMMARY



REMEMBER

Broadband = High speed + Always-on + Wideband
It connects the world faster!



<https://www.shishirbhattarai.com.np>

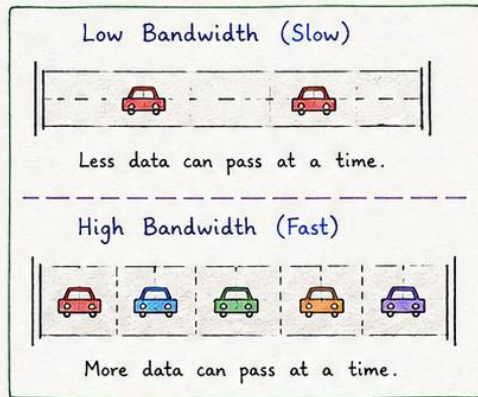
BANDWIDTH

Bandwidth is the maximum amount of data that can be transmitted over a communication channel in a given time.

1. SIMPLE IDEA

Think of bandwidth like a road.

- Narrow road → few cars → slow traffic
- Wide road → many cars → fast traffic



2. PIPE ANALOGY

Bandwidth is like the width of a pipe.

Narrow pipe → less water



Wide pipe → more water



More bandwidth =
More data can flow
at the same time.

3. BANDWIDTH vs SPEED

- Bandwidth → How much data can be carried.
- Speed → How fast the data is actually transferred.

High bandwidth usually means better speed, but not always. (It depends on network congestion, devices, distance, etc.)

4. EXAMPLES (APPROX.)

	1 - 5 Mbps	→ Basic browsing, Emails
	10 - 50 Mbps	→ HD video streaming
	100+ Mbps	→ Online gaming, 4K streaming, Multiple devices

5. BANDWIDTH IS MEASURED IN

- bps → bits per second
- Kbps → Kilobits per second
- Mbps → Megabits per second
- Gbps → Gigabits per second

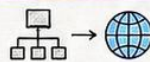
1 Mbps
= 1,000 Kbps
= 1,000,000 bps

6. LOW vs HIGH BANDWIDTH

	LOW BANDWIDTH	HIGH BANDWIDTH
Data Capacity	Less data at a time	More data at a time
Speed	Slow	Fast
Performance	Buffering, delays	Smooth, real-time
Use	Basic tasks	HD streaming, gaming, video calls, large downloads

7. TYPES OF BANDWIDTH

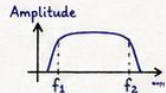
① Network Bandwidth → Capacity of a network connection (Internet, LAN, etc.)



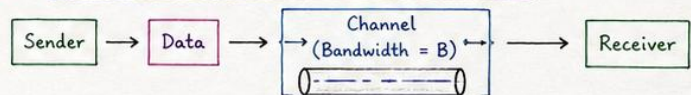
② Digital Bandwidth → Data rate in digital communication systems.



③ Analog Bandwidth → Range of frequencies that an analog signal can carry.

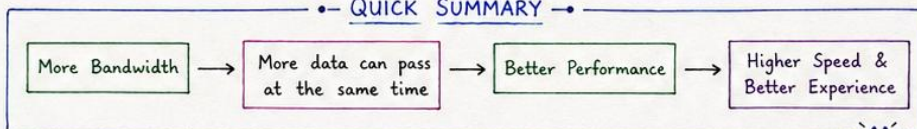


8. BANDWIDTH IN COMMUNICATION CHANNEL



If data rate (R) ≤ Bandwidth (B) → Smooth transmission
If data rate (R) > Bandwidth (B) → Congestion, data loss

QUICK SUMMARY



REMEMBER

Bandwidth is the data carrying capacity of a channel.
Wider the bandwidth,
Better the communication!



3G / 4G / 5G

3G, 4G and 5G are **generations** of mobile communication technology.
Each generation improves speed, capacity, latency and overall performance.

1. 3G (Third Generation)

First generation to properly support mobile internet.

FEATURES

- Internet browsing
- Email, basic apps
- Video calling (limited)
- Music & online services



SPEED : 0.5 – 2 Mbps

GOOD FOR :

- Basic browsing
- Social media (light use)
- Emails

2. 4G (Fourth Generation)

Major upgrade with high-speed internet.

FEATURES

- HD video streaming
- Online gaming
- Video calls
- Smooth apps (YouTube, Netflix, etc.)



SPEED : 10 – 100 Mbps

GOOD FOR :

- Streaming videos
- Video calls
- Fast downloads
- Online gaming

3. 5G (Fifth Generation)

Latest and fastest technology.

FEATURES

- Ultra-fast speed
- Very low latency (almost no delay)
- Supports IoT, smart cities, self-driving tech
- Connects massive devices



**SPEED : 1 Gbps+
(can reach up to 10 Gbps)**

GOOD FOR :

- 4K / 8K streaming
- Virtual Reality (VR)
- Smart devices & automation
- IoT applications

4. QUICK COMPARISON

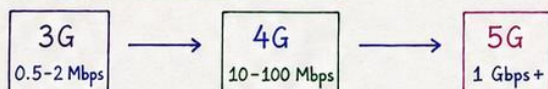
FEATURE	3G	4G	5G
Speed	0.5 – 2 Mbps	10 – 100 Mbps	1 Gbps+
Latency (Delay)	High	Medium	Very Low
Capacity	Low	High	Very High
Usage	Basic web, emails	Streaming, gaming, video calls	IoT, VR, smart cities, self-driving tech
Technology	Early data	Broadband	Advanced, future-ready
Launch (Approx.)	2000	2010	2020 and beyond

5. GENERATION IMPROVEMENT

- 3G** → Internet starts (Mobile connectivity)
- 4G** → Fast internet (Better speed & experience)
- 5G** → Ultra-fast & smart (Future technologies)



6. HOW SPEED IMPROVES



Speed increases → Experience improves → Possibilities grow

7. KEY POINTS

- 3G → Basic internet
- 4G → High speed, better connectivity
- 5G → Ultra-fast, low latency, supports future tech
- Each generation is backward compatible.



★ In One Line : 3G made it possible, 4G made it fast, and 5G will make it limitless! 😊

DATA PACKETS

Data packets are small units of data into which a large message is divided before transmission over a network.

1. WHAT IS A DATA PACKET?

A data packet is a formatted unit of data that carries both control information (header) and user data (payload).

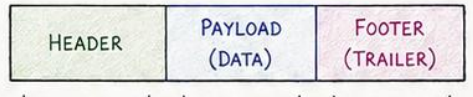


KEY POINTS

- Breaks large data into smaller parts.
- Each packet travels independently.
- Packets may take different routes.
- Receiver reassembles packets to get the original message.



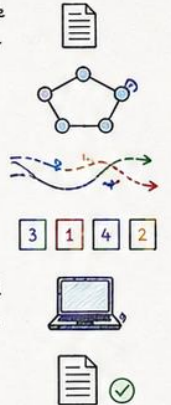
2. STRUCTURE OF A DATA PACKET



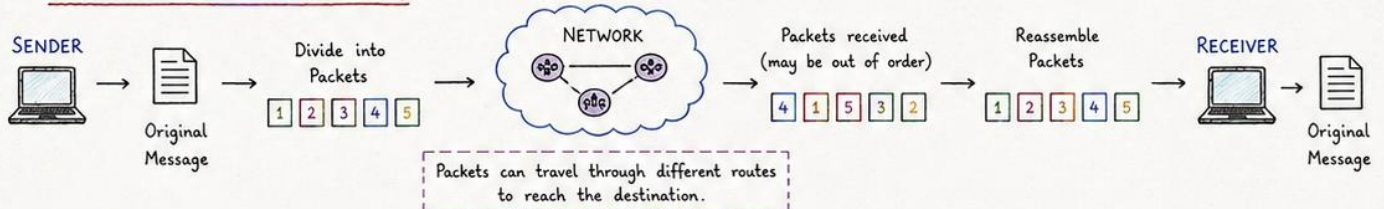
- Source address
 - Destination address
 - Packet sequence number
 - Control information
- Actual data (a part of the original message)
- Error detection (CRC / Checksum)
 - Ensures data integrity

3. HOW DATA PACKETS WORK

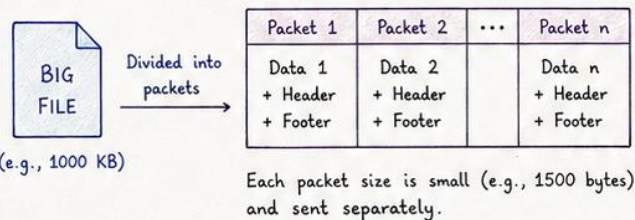
- ① The sender breaks the big data into packets.
- ② Each packet is sent over the network.
- ③ Packets may take different paths.
- ④ Packets may arrive out of order.
- ⑤ Receiver reorders and reassembles packets.
- ⑥ Original message is recovered.



4. TRANSMISSION PROCESS (EXAMPLE)

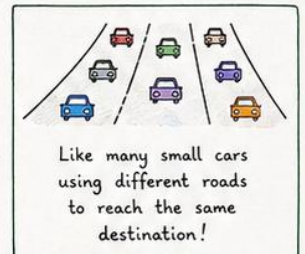


5. EXAMPLE (SENDING A BIG FILE)

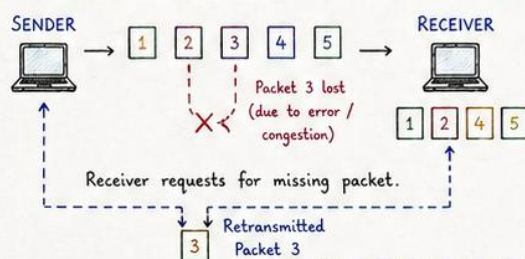


6. ADVANTAGES OF USING PACKETS

- ✓ Efficient use of network bandwidth.
- ✓ Multiple users can share the network.
- ✓ Packets can take alternate routes.
- ✓ Error detection and retransmission is easy.
- ✓ Large data can be transmitted easily.
- ✓ Reliable and scalable communication.



7. PACKET LOSS & RETRANSMISSION

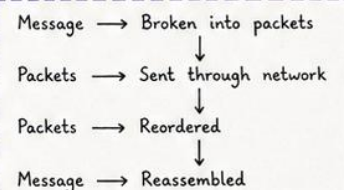


8. COMMON PROTOCOLS THAT USE PACKETS

- IP (Internet Protocol)
- TCP (Transmission Control Protocol)
- UDP (User Datagram Protocol)
- ICMP (Internet Control Message Protocol)
- HTTP / HTTPS
- FTP, SMTP, DNS, etc.



9. QUICK SUMMARY



REMEMBER
Packets are small parts of a big message that travel across the network and are reassembled at the destination to form the original message.



FREQUENCY

Frequency is the number of times a signal (wave) repeats or completes one cycle in one second.

1. WHAT IS FREQUENCY?

It tells us how fast a wave oscillates or vibrates.

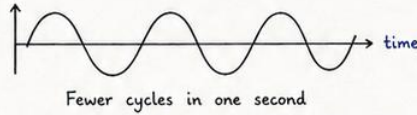
KEY POINTS

- Measured in Hertz (Hz)
- 1 Hz = 1 cycle per second
- Higher frequency → more cycles in a second
- Lower frequency → fewer cycles in a second

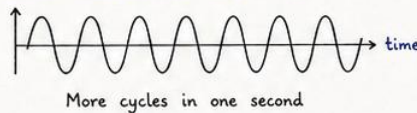


2. WAVE DIAGRAM

LOW FREQUENCY (Low Hz)



HIGH FREQUENCY (High Hz)



3. UNIT OF FREQUENCY

Unit : Hertz (Hz)

1 Hz = 1 cycle per second

EXAMPLES

- 1 Hz → 1 cycle per second
- 10 Hz → 10 cycles per second
- 1 kHz → 1,000 cycles per second
- 1 MHz → 1,000,000 cycles per second
- 1 GHz → 1,000,000,000 cycles/sec

4. FORMULA

Frequency (f) is the reciprocal of Time Period (T)

$$f = \frac{1}{T}$$

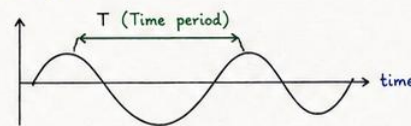
Where,

f = Frequency (in Hz)

T = Time period of one cycle (in seconds)

5. TIME PERIOD (T)

Time period is the time taken to complete one cycle.

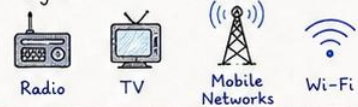


If a wave completes 1 cycle in T seconds,

$$\text{then } f = \frac{1}{T} \text{ Hz}$$

6. FREQUENCY IN COMMUNICATION

- Different frequencies are used to carry different signals.
- Used in radio, TV, mobile networks, Wi-Fi, satellites, etc.
- Helps avoid interference between signals.



Example : Radio stations use different frequencies (e.g., 98 MHz, 102 MHz)

7. FREQUENCY BANDS

Band Type	Frequency Range (Approx.)	Features
Low Frequency (LF)	30 kHz - 300 kHz	Long range, can penetrate walls and water, low data rate
Medium Frequency (MF)	300 kHz - 3 MHz	AM Radio broadcasting
High Frequency (HF)	3 MHz - 30 MHz	Long range (ionosphere reflection)
Very High Frequency (VHF)	30 MHz - 300 MHz	FM Radio, TV broadcasting
Ultra High Frequency (UHF)	300 MHz - 3 GHz	TV, Mobile, Wi-Fi, Bluetooth
Super High Frequency (SHF)	3 GHz - 30 GHz	Satellites, 4G, 5G, Wi-Fi

8. EFFECT OF FREQUENCY

LOW FREQUENCY

- Longer wavelength
- Travels longer distance
- Can pass through obstacles
- Lower data capacity



HIGH FREQUENCY

- Shorter wavelength
- Travels shorter distance
- Blocked by obstacles
- Higher data capacity



9. QUICK COMPARISON

Feature	Low Frequency	High Frequency
Wavelength	Long	Short
Range	Long	Short
Obstacle Penetration	High	Low
Data Capacity	Low	High
Examples	AM Radio	Wi-Fi, 5G

IN ONE LINE

Frequency tells us how many times a wave repeats in one second.

Higher the frequency, more cycles per second!



10. REAL-LIFE EXAMPLES

AM Radio (0.53 - 1.6 MHz)	FM Radio (88 - 108 MHz)	TV Broadcasting (VHF : 30 - 300 MHz) (UHF : 300 MHz - 3 GHz)	Mobile Networks (700 MHz - 2.6 GHz)	Wi-Fi (2.4 GHz / 5 GHz)	5G Networks (3 - 30 GHz)
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CAT6 CABLE vs FIBER OPTIC CABLE

Both are **wired communication media** used to transmit data.
They differ in technology, speed, distance and performance.

1. CAT6 CABLE (Category 6)

CAT6 is a type of **twisted pair copper cable** used in Ethernet networks.

HOW IT WORKS

Uses **electrical signals** to transmit data.
Contains twisted copper wires to reduce interference.



FEATURES

- Speed : up to 1 Gbps (1000 Mbps)
- Max speed (short distance) : up to 10 Gbps (≈ 55 m)
- Distance : up to 100 meters
- Cost : cheaper than fiber

ADVANTAGES

- Easy to install
- Affordable
- Widely used in homes / offices

DISADVANTAGES

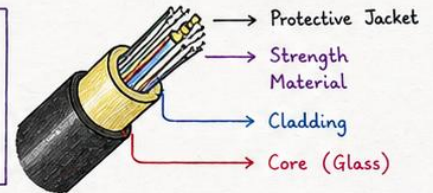
- Affected by electromagnetic interference (EMI)
- Lower speed compared to fiber
- Limited distance (max 100 m)

2. FIBER OPTIC CABLE

Fiber optic cable uses **light signals** instead of electricity.

HOW IT WORKS

Data is transmitted as **light pulses** through glass or plastic fibers.



FEATURES

- Speed : 10 Gbps to 100+ Gbps
- Distance : kilometers without signal loss
- Extremely low latency
- Signal type : Light (optical)

ADVANTAGES

- Ultra-fast speed
- No interference (immune to EMI)
- Long-distance transmission
- More secure

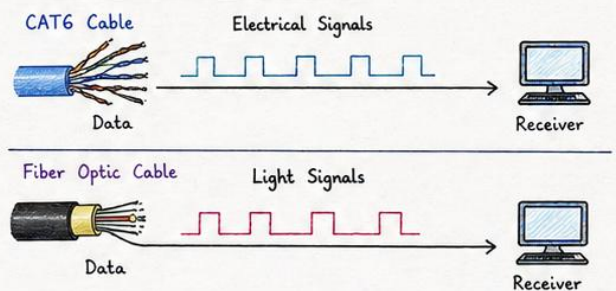
DISADVANTAGES

- Expensive
- Difficult to install / repair
- More fragile than copper cables

3. QUICK COMPARISON

Feature	CAT6 Cable	Fiber Optic Cable
Signal Type	Electrical	Light
Speed	Up to 1-10 Gbps	10-100+ Gbps
Distance	Up to 100 m	Kilometers
Interference	Yes (affected by EMI)	No (immune to EMI)
Latency	Higher	Very low
Cost	Low	High
Installation	Easy	Difficult
Durability	More robust	More fragile

4. HOW THEY TRANSMIT DATA



5. WHERE ARE THEY USED ? (EXAMPLES)

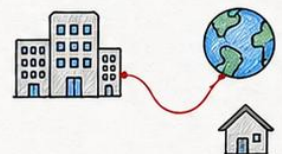
CAT6 CABLE

- Home networks
- Office LAN (Local Area Network)
- Connect computers, printers, routers, switches, etc.



FIBER OPTIC CABLE

- Internet backbone
- Long-distance communication
- Data centers
- Telecom networks, FTTH (Fiber to Home)
- High-speed internet, TV, 5G backhaul



IN ONE LINE : CAT6 is a copper cable for short-distance, cost-effective networks.

Fiber optic is a light-based cable for ultra-fast, long-distance, high-performance networks.



WIRELESS COMMUNICATION TECHNOLOGIES

These technologies use **electromagnetic waves** to communicate without any physical (wired) connection.

1. Wi-Fi

Wi-Fi is a wireless technology used to provide internet access to devices.

HOW IT WORKS



Router sends radio waves and devices connect to the internet.

FEATURES

- Uses radio waves (2.4 GHz / 5 GHz / 6 GHz)
- Range : ~20 - 100 meters
- Speed : High (tens of Mbps to Gbps)
- Supports multiple devices

USES

- Internet on phones, laptops
- Home / office networks
- Online gaming, streaming, etc.

2. Bluetooth

Bluetooth is used for short-range communication between devices.

HOW IT WORKS



Devices pair with each other using short-range radio waves.

FEATURES

- Uses low-power radio waves (2.4 GHz)
- Range : ~10 - 50 meters
- Speed : Low to Moderate
- Low power consumption
- Connects a few devices at a time

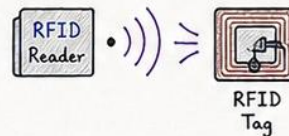
USES

- Wireless headphones / speakers
- File transfer between devices
- Keyboards, mouse, printers
- Smart watches, fitness bands

3. RFID

RFID (Radio Frequency Identification) is used to identify and track objects using radio waves.

HOW IT WORKS



1. Reader sends radio signal.
2. Tag receives the signal.
3. Tag sends back data to reader.

FEATURES

- Uses radio frequency waves
- No need for line of sight
- Works at short distances (cm to a few meters)
- Can read multiple tags quickly

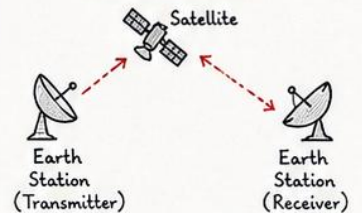
USES

- ID cards and access control
- Inventory / asset tracking
- Library books, toll collection
- Supply chain management

4. Satellites

Satellites enable long-distance communication across the globe.

HOW IT WORKS



Signals are sent from Earth station to satellite and back to another location.

FEATURES

- Very large coverage (global)
- Long distance communication
- Works in remote areas
- High cost of setup
- Some delay (latency)

USES

- TV broadcasting (DTH)
- GPS navigation
- Weather monitoring
- Internet in remote areas

5. QUICK COMPARISON

Feature	Wi-Fi	Bluetooth	RFID	Satellites
Type	Wireless LAN	Personal Area Network (PAN)	Identification System	Wide Area Coverage
Range	Medium (20 - 100 m)	Short (10 - 50 m)	Very Short (cm - few m)	Very Long (Global)
Speed	High (Up to Gbps)	Low to Moderate (Up to few Mbps)	Very Low (Kbps)	High (Depends on link)
Power Use	Medium	Very Low	Very Low (Tag has low power)	High
Main Use	Internet Access	Device Connectivity	Identification & Tracking	Broadcasting & Global Comm.

6. KEY POINTS

- ★ Wi-Fi → High speed internet over medium range.
- ★ Bluetooth → Short range, low power device connection.
- ★ RFID → Identify and track objects using radio waves.
- ★ Satellites → Provide global communication over long distances.



IN ONE LINE : Wi-Fi connects us to the internet, Bluetooth connects devices, RFID identifies things, and Satellites connect the world !



CONNECTORS & DEVICES : RJ45 AND MEDIA CONVERTER

These are important components used in network communication systems.

1. RJ45 CONNECTOR

RJ45 (Registered Jack 45) is a standard connector used for Ethernet networking.

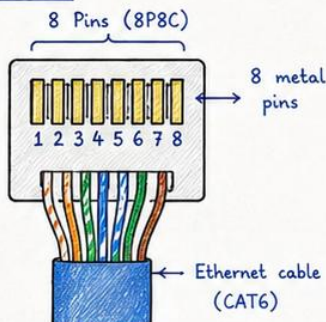
WHAT IT DOES

- Connects network cables (like CAT6) to devices.
- Used to transmit **electrical signals**.

WHERE YOU SEE IT

- Computers 
- Routers 
- Switches 
- Modems 

STRUCTURE

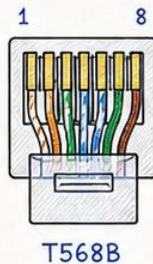


FEATURES

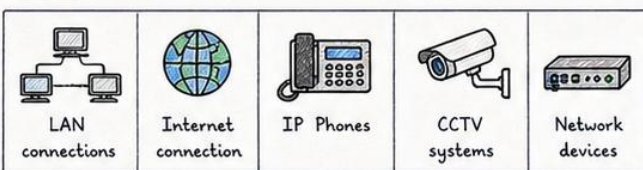
- Supports Ethernet networks (LAN)
- Easy to plug and unplug
- Works with CAT5e, CAT6 cables
- 8P8C - 8 Positions, 8 Contacts

PINOUT (T568B - Most Common)

Pin No.	Wire Color	Function
1	White-Orange	Transmit Data +
2	Orange	Transmit Data -
3	White-Green	Receive Data +
4	Blue	(Not used)
5	White-Blue	(Not used)
6	Green	Receive Data -
7	White-Brown	(Not used)
8	Brown	(Not used)



USES



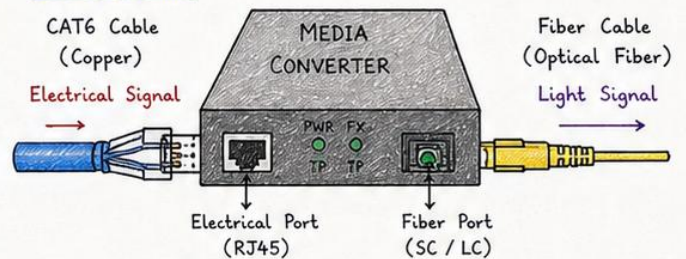
2. MEDIA CONVERTER

A media converter is a device that converts one type of communication media into another.

WHAT IT DOES

- Converts one medium to another.
- Most common use:
Copper (Ethernet) ↔ Fiber Optic

HOW IT WORKS



IT CONVERTS

- Electrical signal → Light signal (Copper to Fiber)
- Light signal → Electrical signal (Fiber to Copper)

WHY IT IS USED

- Extend network over long distances
- Connect different types of cables
- Upgrade from copper to fiber
- Reduce signal loss and interference

FEATURES

- Supports long-distance communication
- Improves network flexibility
- High speed and stable
- Easy to install
- Used in enterprise networks, ISPs, campuses, etc.

QUICK COMPARISON

Feature	RJ45 Connector	Media Converter
Type	Connector	Device
Function	Connects cables	Converts signal / media type
Signal	Electrical	Electrical ↔ Light
Usage	LAN connections	Fiber-copper integration
Form	Small plug	Standalone box
Example	RJ45 plug on network cable	Device between switch and fiber cable



IN ONE LINE : RJ45 is the plug that connects network cables.

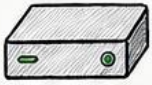
Media converter is the device that converts **copper** signals to **fiber** signals (and vice versa).



📶 NETWORKING DEVICES 📶

These devices are used to **connect** devices and networks and to **transfer data** efficiently.

1. REPEATER



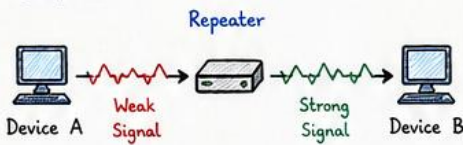
Repeater

The simplest device. It regenerates (boosts) weak signals and **extends** the network distance.

KEY POINTS

- Does not filter data.
- Just **repeats** the signal.

DIAGRAM :



USED IN : Long cable networks to extend distance.

2. HUB



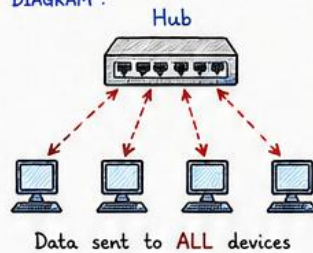
Hub

It connects multiple devices in a network. It sends data to **all** connected devices.

KEY POINTS

- No intelligence.
- **Broadcasts** data to all ports.
- Can cause network traffic.

DIAGRAM :



WORKS AT : Physical Layer (Layer 1)

3. SWITCH



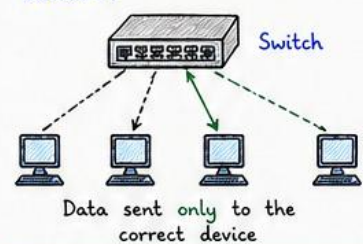
Switch

It is an improved version of hub. It sends data only to the **correct** device.

KEY POINTS

- Uses MAC address.
- Sends data **only** to the destination device.
- Reduces network congestion.

DIAGRAM :



WORKS AT : Data Link Layer (Layer 2)

4. BRIDGE



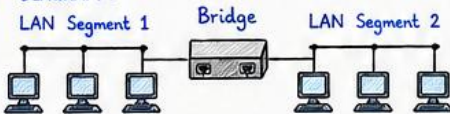
Bridge

It connects two network segments. It **filters** traffic between networks and reduces unnecessary data flow.

KEY POINTS

- Uses MAC address.
- Forwards data only when needed.
- Smarter than Hub.

DIAGRAM :



WORKS AT : Data Link Layer (Layer 2)

5. ROUTER



Router

It connects different networks (like LAN to Internet). It routes data using **IP address**.

KEY POINTS

- Uses IP address.
- Chooses the best path for data.
- Most intelligent device here.

DIAGRAM :



WORKS AT : Network Layer (Layer 3)

SUMMARY (IN ONE LINE)

- Repeater → Boosts (Regenerates) signal.
- Hub → Sends data to all devices.
- Switch → Sends data to **correct** device.
- Bridge → Connects two LANs.
- Router → Connects different networks.

📶 QUICK COMPARISON 📶

Feature	Repeater	Hub	Switch	Bridge	Router
Function	Boosts weak signal	Broadcasts data to all ports	Sends data to specific device	Connects two LAN segments	Connects different networks
Intelligence	None	Low	Medium	Medium	High
Address Used	—	—	MAC Address	MAC Address	IP Address
OSi Layer	Physical Layer (Layer 1)	Physical Layer (Layer 1)	Data Link Layer (Layer 2)	Data Link Layer (Layer 2)	Network Layer (Layer 3)
Data Forwarding	Repeats signal	To all devices	To one device (destination)	Filters and forwards when needed	Routes using best path
Example Use	Long distance cable networks	Small networks (not recommended)	Most LANs (offices, homes)	Connecting two LANs	Internet connection, inter-networking



REMEMBER : Repeater (Boost) → Hub (To All) → Switch (To One) → Bridge (Connect LANs) → Router (Connect Networks)



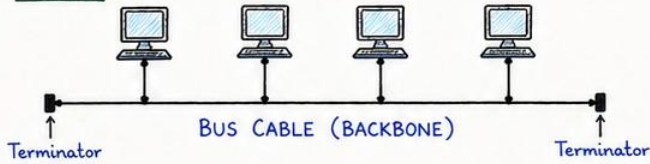
📶 NETWORK TOPOLOGIES OVERVIEW 📶

A network topology is the **physical** or **logical** layout of how devices are connected in a network.

1. BUS TOPOLOGY

Concept : All devices are connected to a single main cable (backbone).

DIAGRAM :



ADVANTAGES

- Simple and low cost
- Easy to install
- Uses less cable

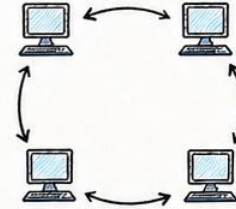
DISADVANTAGES

- If main cable fails → entire network down
- Data collisions can occur
- Difficult to detect faults

2. RING TOPOLOGY

Concept : Devices are connected in a circular loop.

DIAGRAM :



ADVANTAGES

- No data collision
- Equal access for all devices
- Data flows in one direction

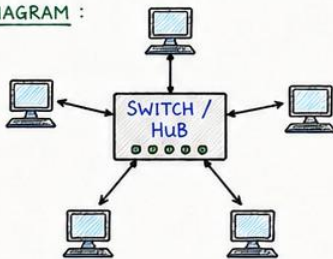
DISADVANTAGES

- One device failure can affect entire network
- Harder to troubleshoot
- Adding/removing devices can break the ring

3. STAR TOPOLOGY

Concept : All devices connect to a central device (hub/switch).

DIAGRAM :



ADVANTAGES

- Easy to manage
- Easy to add/remove devices
- Failure of one device doesn't affect others
- Good performance

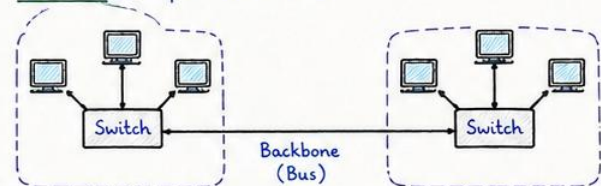
DISADVANTAGES

- If central device fails → entire network down
- More cable required
- Cost is higher than bus

4. HYBRID TOPOLOGY

Concept : Combination of two or more topologies.

DIAGRAM (Example : Star + Bus)



ADVANTAGES

- Flexible and scalable
- Reliable
- Easy to expand

DISADVANTAGES

- Complex design
- Expensive
- Installation & maintenance is difficult

QUICK COMPARISON

Topology	Structure	Reliability	Cost	Complexity	Cable Usage	Best Suited For
Bus	Single backbone cable	Low	Low	Simple	Less	Small, temporary networks
Ring	Circular loop	Medium	Medium	Moderate	Medium	Networks needing equal access
Star	Central hub/switch	High	Medium	Easy	More	Most LANs (Offices, Labs)
Hybrid	Combination of topologies	Very High	High	Complex	More	Large & complex networks



IN ONE LINE : Bus → One cable for all, Ring → Circular connection,
Star → Central hub/switch, Hybrid → Combination of topologies.



<https://www.shishirbhattarai.com.np>

📶 NETWORKS BASED ON COVERAGE 📶

Networks can be classified based on the **geographical area** they cover - from very small to **global**.

1. PAN

(Personal Area Network)

CONCEPT

A very small network around a single person.

COVERAGE

Few meters
($\approx 1 - 10$ m)

EXAMPLES

- Bluetooth devices
- Smartwatch
- Phone ↔ Laptop connection



KEY POINT

Short range, low power.



Very Small Area

2. LAN

(Local Area Network)

CONCEPT

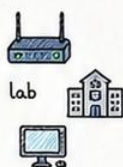
Network within a small area like home, school or office.

COVERAGE

Up to a few kilometers.

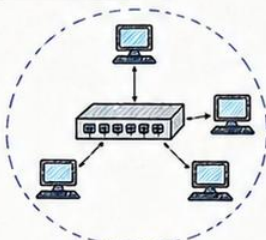
EXAMPLES

- Home Wi-Fi
- School computer lab
- Office network



KEY POINT

High speed, privately owned.



Small Area

3. MAN

(Metropolitan Area Network)

CONCEPT

Network covering a city or large campus area.

COVERAGE

Several kilometers
(city - wide)

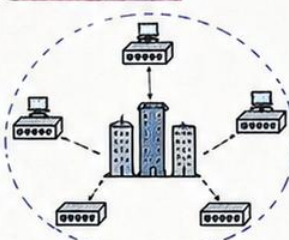
EXAMPLES

- City internet services
- Cable TV networks



KEY POINT

Connects multiple LANs.



City Area

4. WAN

(Wide Area Network)

CONCEPT

Network covering large regions, countries or the world.

COVERAGE

Thousands of kilometers.

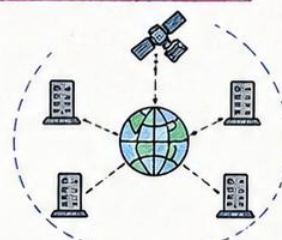
EXAMPLES

- Internet
- Bank networks
- Satellite communication



KEY POINT

Uses public networks and advanced infrastructure.



Global Area

📶 QUICK COMPARISON 📶

Feature	PAN	LAN	MAN	WAN
Full Form	Personal Area Network	Local Area Network	Metropolitan Area Network	Wide Area Network
Coverage	Few meters (1 - 10 m)	Up to a few kilometers	Several kilometers (city-wide)	Thousands of kilometers
Area	Very small	Small (Home/Office)	City or large campus	Country to Global
Speed	Low	High	Medium	Varies (usually lower)
Ownership	Personal	Private	Private / Public	Public
Example	Bluetooth, Smartwatch	Wi-Fi, Office Network	City Internet, Cable TV	Internet, Bank Network



IN ONE LINE : PAN is Personal, LAN is Local, MAN is Metropolitan, WAN is Wide.



NETWORK ARCHITECTURE

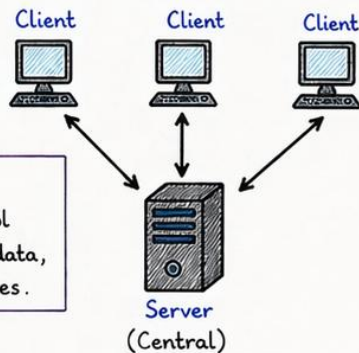
Network architecture defines how devices are **organized** and how they **communicate** in a network.

1. CLIENT-SERVER ARCHITECTURE

CONCEPT :

A central **server** provides services, and multiple **clients** request those services.

DIAGRAM :



FEATURES

- Centralized control
- Server manages data, security, resources.

ADVANTAGES

- Better security
- Centralized data management
- Easy backup

DISADVANTAGES

- Expensive (needs powerful server)
- If server fails → whole network affected

EXAMPLES

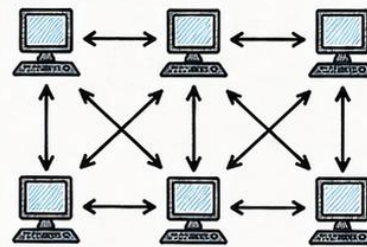


2. PEER-to-PEER (P2P) ARCHITECTURE

CONCEPT :

All devices (**peers**) are equal and share resources directly with each other.

DIAGRAM :



FEATURES

- No central server
- Each device acts as both client and server.

ADVANTAGES

- Low cost
- Easy to set up
- No central dependency

DISADVANTAGES

- Less secure
- Hard to manage
- Data not centralized

EXAMPLES



QUICK COMPARISON

Feature	Client-Server Architecture	Peer-to-Peer (P2P) Architecture
Control	Centralized	Decentralized
Cost	High	Low
Security	High	Low
Management	Easy	Difficult
Data Storage	Centralized on server	Distributed on all peers
Dependency	Server dependent	No central device
Best For	Large organizations, services	Small networks, file sharing

IN ONE LINE

★ **Client-Server** → Central server controls everything.

★ **Peer-to-Peer** → All devices share equally.

